

Punya Srivalli Nallamothu

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EDUCATION

University of Texas at Dallas

Masters in Business Analytics and AI **CGPA : 4.0/4.0**

Dallas, Texas

August 2023 - May 2022

BVRIT Hyderabad College of Engineering for Women

Bachelor of Electrical and Electronics Engineering **CGPA : 7.86/10**

Hyderabad, India

June 2018 - July 2022

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, R, Java, C, HiveQL
- **Libraries/Frameworks:** Pandas, NumPy, Scikit-learn, Keras, PyTorch, PySpark, Hive, Impala, Apache Spark, Sqoop, Django, SciPy
- **Database:** MySQL, Hadoop Distributed File System, PostgreSQL
- **Software Tools:** AWS (Glue, Redshift, S3, Sagemaker), GCP (BigQuery), Snowflake, Git, Azure, Docker, Tableau, PowerBI, JIRA, Hugging Face, Microsoft Excel, Airflow
- **Coursework:** Predictive & Prescriptive Analytics, Big Data, Applied Econometrics & Time Series Analysis, Advanced Statistics, Data Visualization, Database Management Systems, Machine Learning, Deep Learning

WORK EXPERIENCE

University of Texas at Dallas

Dallas, Texas

Student Assistant

December 2023 – May 2024

- Designed and developed a web-based Digital Asset Management system using **Django** framework and **MySQL** to organize and retrieve image/video content for course development; implemented a dynamic tagging and filtering UI based on scenario, location, action, and persona type - streamlining asset discovery and improving content reuse efficiency for the client
- Assisted in deploying learning materials on LearnWorlds, including role-based access management, course segmentation, and format conversion - enhancing scalability of training programs and enabling personalized learning experiences for associates
- Conducted cost-benefit analysis across 4+ SaaS tools, driving cost savings and informing data-driven decisions on technology investments and tool consolidation

Accenture

Hyderabad, India

Custom Software Engineering Associate

October 2022 – June 2023

- Developed and customized master data management solution using Java and **TIBCO EBX** for healthcare data of a pharmaceutical company, enabling insertions and cascading deletions of provider records across interconnected tables, ensuring data integrity and consistency across systems
- Built robust data pipelines using **Spring Batch** to transform and standardize streaming data from 3 vendors, facilitating accurate and timely integration into the master data repository and improving downstream analytics reliability

RESEARCH PROJECTS

Health Cost Navigator

January 2025 – May 2025

- Built an end-to-end ETL pipeline using **Python** to scrape, parse, and standardize healthcare pricing data from hospital websites across 8 U.S. cities, ensuring consistency across multiple insurance providers
- Designed a Star Schema-based data warehouse and transformed healthcare cost datasets in **Snowflake** using SQL, improving query performance and enabling scalable, analytics-ready data storage for downstream BI tools
- Implemented batch processing for high-volume datasets, reducing Snowflake compute costs and improving data pipeline efficiency
- Developed interactive dashboards in **Streamlit** and **PowerBI**, allowing users to compare medical service costs across hospitals and insurance plans - promoting price transparency and informed decision-making

Uber and Lyft Ride Price Prediction

September 2024 – December 2024

- Utilized 5 predictive models, including **Random Forest Regressor** and **Decision Tree Regressor**, to analyze complex relationships between 50+ input variables and ride prices. Compared model performance metrics, with the best-performing model achieving a 96% R-squared value and an RMSE of 1.79
- Performed a classification task using supervised learning algorithms, including Random Forest and **XGBoost**, to categorize ride prices into 6 bins. The best-performing model achieved an F1 score of 0.85, effectively predicting price ranges and supporting targeted pricing strategies
- Carried out exploratory data analysis (**EDA**) and feature engineering to enhance model inputs. Applied hypothesis testing and **ANOVA** to validate feature significance, resulting in improved model accuracy

ANT'S Truck Risk Factor Analysis

March 2024 – May 2024

- Processed 3 years of telematics data (including GPS coordinates, truck events, and driver behavior) using Apache Pig and PySpark on Hadoop to calculate driver-specific risk scores, improving risk visibility across 300+ fleet assets
- Identified high-risk behaviors like unsafe following distance, speeding and flagged 10% of drivers exceeding the risk threshold (≥ 7.0) for targeted safety interventions
- Built interactive dashboards by integrating **Impala with Tableau and R**, enabling real-time monitoring of driver performance and regional event trends. Mapped event density across U.S. cities to **optimize routing and resource allocation**, justifying adjustments in shipping charges based on location-specific risk

ACHIEVEMENTS AND CERTIFICATIONS

- Ranked in the top 1% of students in the Master's program in Business Analytics and AI at UTD, showcasing academic excellence.
- Achieved AWS Certified Cloud Practitioner.
- Secured fourth place in a national-level technical hackathon, focusing on solving environmental challenges using AI.